Ex No: 3a

IMPLEMENTATION OF BLOCKS WORLD PROGRAM

Aim:

To implement binary search tree.

Procedure:

Program:

```
# Class representing the state of the blocks world
class BlocksWorld:
    def __init__(self, num_blocks):
        # Initialize blocks world with num_blocks on the table
        self.state = [[block] for block in range(num_blocks)]  # Each block starts on the table
        self.num_blocks = num_blocks

    def display_state(self):
        # Display the current state of the blocks world
        for stack in self.state:
            print(f"Block(s) on stack: {stack}")

    def move(self, block, destination):
        # Move block onto another block or the table
        # Find the block in the current state and move it
        source_stack = self.find_block(block)
```

```
        destination_stack = self.find_block(destination)

        if source_stack is None or destination_stack is None:
            print(f"Invalid block {block} or destination {destination}.")
            return

        # Remove the block from its current position
        source_stack.remove(block)

        # Put the block on the destination stack
        destination_stack.append(block)

        # Display the state after the move
        self.display_state()

    def find_block(self, block):
        # Find the stack that contains the given block
        for stack in self.state:
            if block in stack:
                return stack
        return None

    def goal_state(self, goal):
        # Goal state is a list of stacks that represents the desired configuration
        self.state = goal
        print("Goal state set.")
        self.display_state()

# Main program to solve Blocks World problem
def main():
    # Initialize the Blocks World with 3 blocks (0, 1, 2)
```

```
    blocks_world = BlocksWorld(3)

    print("Initial state:")
    blocks_world.display_state()

    # Define the goal state (Block 0 is on top of Block 1, and Block 2 is on the table)
    goal = [[0, 1], [2]]  # Goal configuration
    blocks_world.goal_state(goal)

    print("\nPerforming Moves:")
    # Move Block 0 onto Block 2
    blocks_world.move(0, 2)

    # Move Block 1 onto Block 2
    blocks_world.move(1, 2)

    # Move Block 2 onto the table (empty stack)
    blocks_world.move(2, 0)

# Run the main program
if __name___== "__main__":
    main()
```

Output:

Initial state:

Block(s) on stack: [0]

Block(s) on stack: [1]

Block(s) on stack: [2]

Goal state set.

Block(s) on stack: [0, 1]

Block(s) on stack: [2]

Performing Moves:

Block(s) on stack: [1]

Block(s) on stack: [2, 0]

Block(s) on stack: []

Block(s) on stack: [1]

Block(s) on stack: [2, 0]

Block(s) on stack: []

Block(s) on stack: [1, 2]

Block(s) on stack: [0]

Block(s) on stack: []

Terminal - Ex 3a: Implementation of Blocks World Program

```
$ py code/ex3a.py

Initial state:
Block(s) on stack: [0]
Block(s) on stack: [1]
Block(s) on stack: [2]
Goal state set.
Block(s) on stack: [0, 1]
Block(s) on stack: [2]

Performing Moves:
Block(s) on stack: [1]
Block(s) on stack: [2, 0]
Block(s) on stack: []
Block(s) on stack: [2, 0, 1]
Block(s) on stack: []
Block(s) on stack: [0, 1, 2]
```